MODELSKA ANALIZA 1

09: STOHASTIČNI POPULACIJSKI MODELI

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FMF, 2024/2025

UVOD

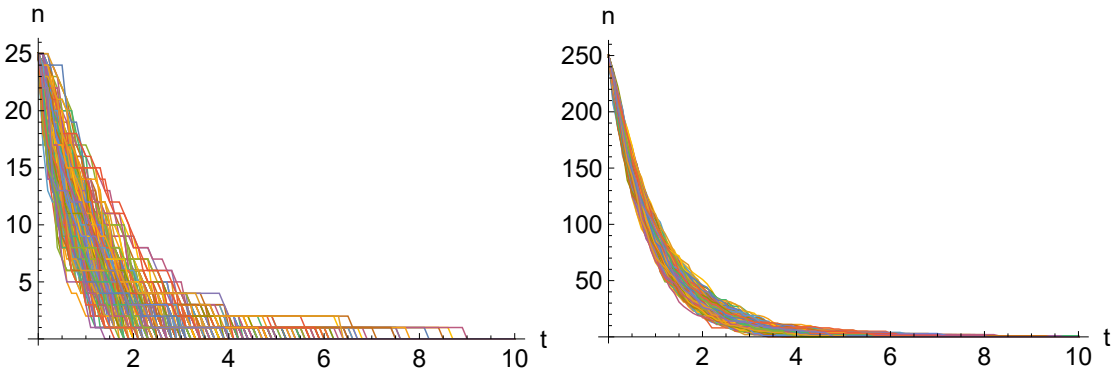
V tej nalogi populacijske modele obravnavamo stohastično. To pomeni, da vse spremembe populacije v času Δt žrebamo po Poisonski porazdelitvi $\frac{\lambda^x e^{-\lambda}}{x!}$.

1.) ČASI IZUMRTJA

Smiselno je, da je hitrost rojstev in smrti sorazmeren s populacijo. To opiše $\dot{n} = (\beta_r - \beta_s)\, n$. V diskretni sliki to nadomesti $\boxed{\Delta n = \mathcal{P}(n\, \beta_r\, \Delta t) - \mathcal{P}(n\, \beta_s\, \Delta t)}$. To v *Mathematici* žrebamo z *RandomInteger@PoissonDistribution[n β Δt]*.

BREZ ROJSTEV

Najprej poglejmo primer, da se sploh ne rojevajo: $\Delta n = -\mathcal{P}(n\, \beta_s\, \Delta t)$. Prikažimo 500 poskusov $n(t)$ za $\beta_s = 1$, $\Delta t = 0.1$ in $n_0 \in \{25, 250\}$.

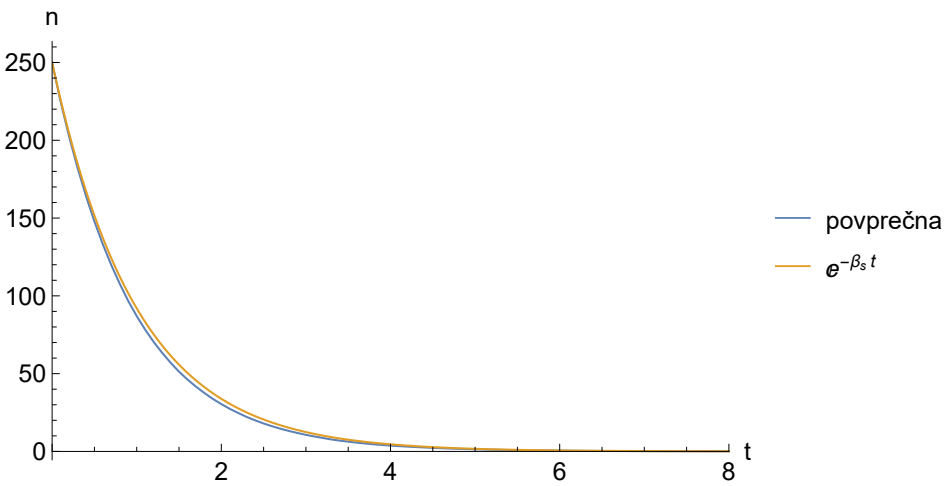


spreminjanje populacije za $n_0 = 25$ in $n_0 = 250$

Vidimo, da takšna stokastična verzija hitro pade na točno 0. To je čas izumrtja.

DODATNO: POVPREČEVANJE STOKASTIČNIH REŠITEV

Postevi se vprašanje: če povprečimo veliko stokastičnih poskusov, ali takšne točke konvergirajo k zvezni rešitvi?



Povprečje stokastičnih poskusov in zvezna rešitev

Iz grafa vidimo da ne konvergiramo k zvezni rešitvi, je pa blizu. Število poskusov je dovolj veliko, da lahko to trdimo, saj graf ni videti zašumljen pa tudi izgleda enako pri 10^3 ali pa 10^6 poskusih.

Kaj pa če pomanjšamo časovni korak na t = .01?

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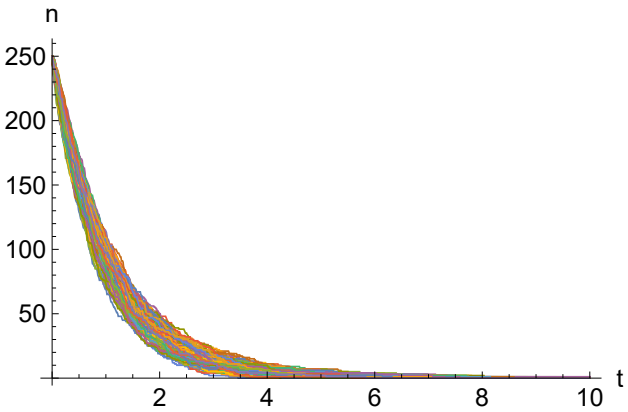

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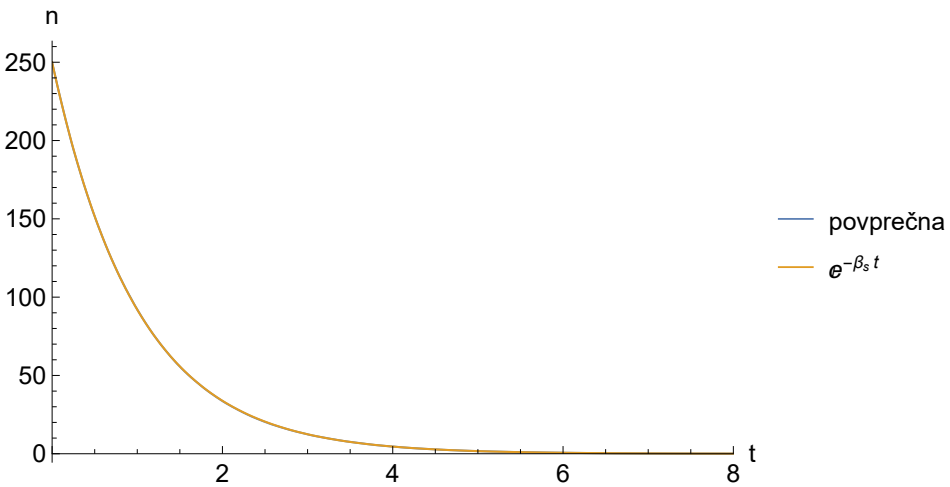

Videti, je, da z zmanjševanjem Δt in povprečenjem pridemo poljubno blizu zvezni rešitvi

PORAZDELITEV ČASA IZUMRTJA

Poglejmo, kako Δt, n_0, β_s vplivajo na porazdelitev časov izumrtja. Osnovne vrednosti so na vseh grafih $\beta_s = 1$, Δt = 0.1, $n_0 = 250$:

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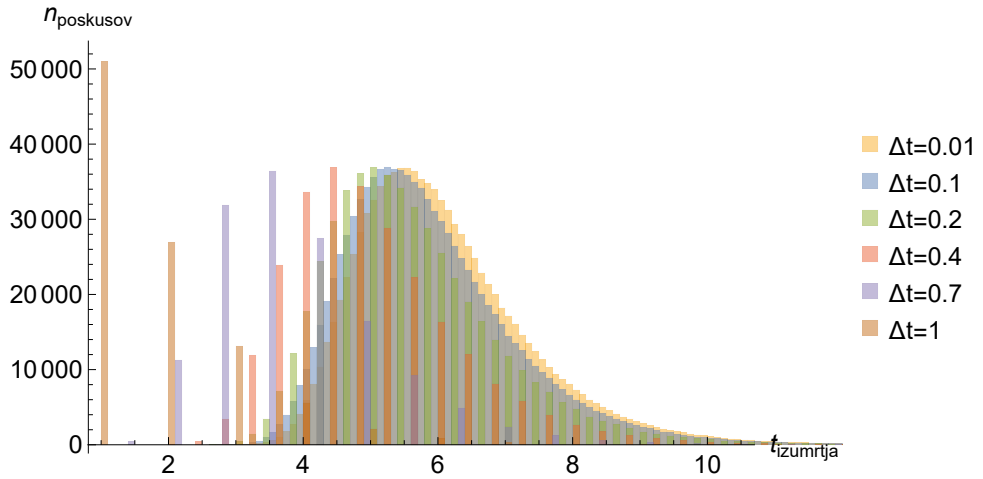

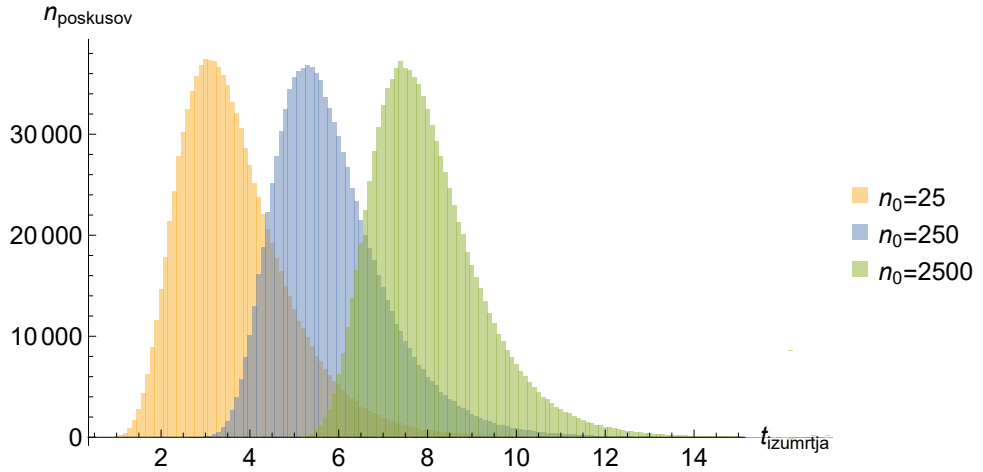

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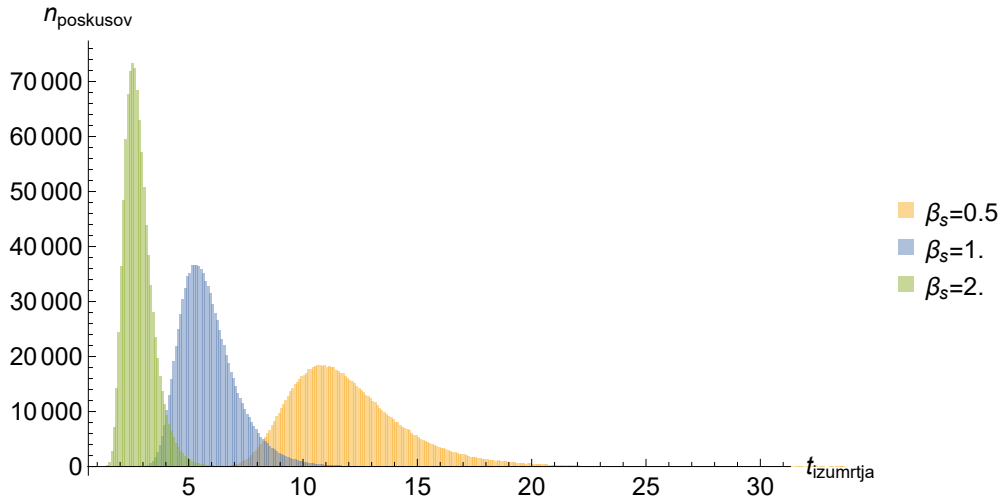
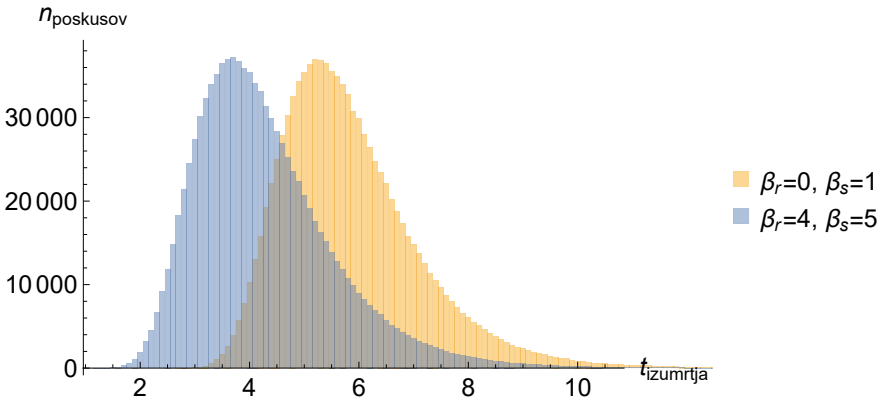


- Opazimo, da Δt vpljia na porazdelitev, a se zdi, da za vse manši in manjši $\Delta t \longrightarrow 0$ porazdelitev časov izumrtja konvergira k neki limitni porazdelitvi.
- Večja začetna populacija samo translira porazdelitev (zakasni izumrtje)
- Za večje β_s se izumrtje zgodi prej in to z manjšo raztresenostjo

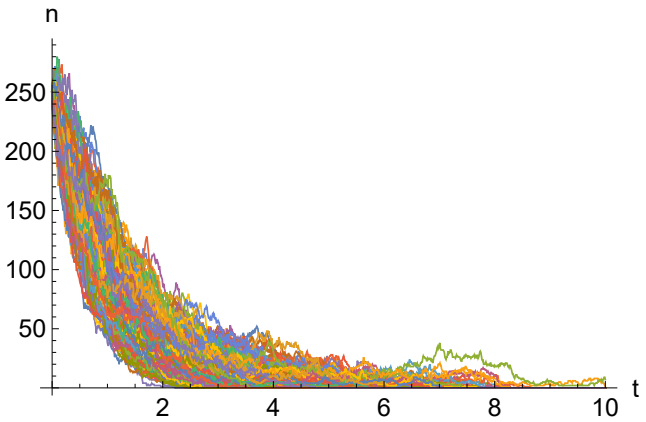
PRIMERJAVA Z UPOŠTEVANJEM ROJSTEV

Vzemimo osnovne parametre $\beta_r = 4$, $\beta_s = 5$, $\Delta t = 0.1$, $n_0 = 250$. Vemo, da mora v zvezni sliki to dati isto rešitev, kot če ni rojstev in je $\beta_s = 1$, $\Delta t = 0.1$, $n_0 = 250$. Vseeno nam to nič ne pove časih izumrtja, poglejmo:



Porazdelitev izumrtji za ustrezajoča primera *z* in brez rojstev

Vidimo, da do izumrtja ride kasneje v primeru z rojstvi. To je podobno, kot smo imeli naključno gibanje levo-desno v absorberski plošči.



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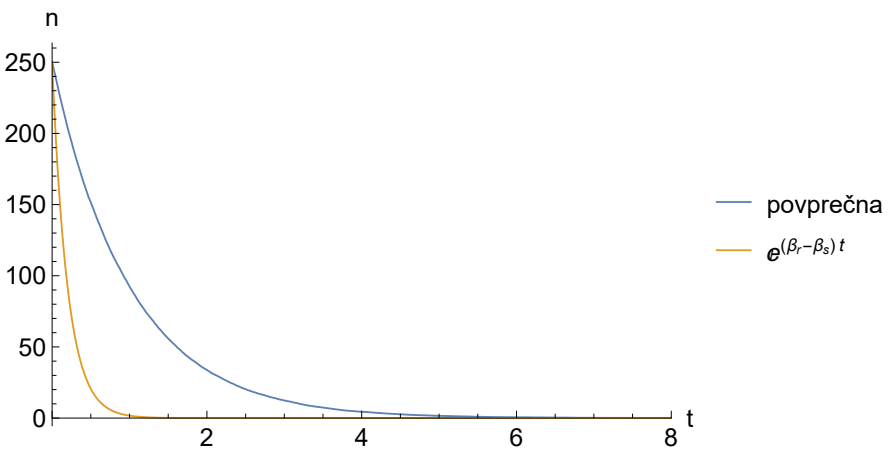

Povprečna rešitev se z manjšanjem Δt, presenetljivo, vse bolj razlikuje od zvezne. Po svoje se zdi logično, saj pride do asimetrije: nad 0 se lahko populacija svobodno naključno sprehaja, pod 0 pa se ne more, zato se težišče premakne navzgor in v desno.

2.) ČASI IZUMRTJA Z MATRIKO PREHODOV

Pravzaprav gre za markovske verige, ki smo jih omenili že pri metropolisu. Imejmo vektor $\boldsymbol{P}$, kjer je $P_i(t)$ verjetnost, da je ob času t populacija enaka $i-1$. Iščemo matriko, za naslednje stanje: $\boldsymbol{P}(t+\Delta t) = \boldsymbol{A}.\boldsymbol{P}(t)$. Element matrike je verjetnost za prehod populacije $n \longrightarrow m$.

- vir: *deepseek R1*
  - $P(m|n) = \sum_{l=0}^{\infty} P(k = m - n + l \,|\, \text{births}) \cdot P(l \,|\, \text{deaths})$

    Where:

    - $P(k \,|\, \text{births}) = \frac{(\lambda_b)^k e^{-\lambda_b}}{k!}$ (Poisson distribution for births)
    - $P(l \,|\, \text{deaths}) = \frac{(\lambda_d)^l e^{-\lambda_d}}{l!}$ (Poisson distribution for deaths)
  - **Step 4: Skellam Distribution**

    The difference between two independent Poisson random variables follows the Skellam distribution. Hence, the transition probability can be expressed using the Skellam probability mass function:

    $$P(m|n) = e^{-(\lambda_b+\lambda_d)} \left(\frac{\lambda_b}{\lambda_d}\right)^{(m-n)/2} I_{m-n}(2\sqrt{\lambda_b \lambda_d})$$

    where:

    - I_k is the modified Bessel function of the first kind.
- Nas pa posebej zanima za majhne čase Δt, kar ustreza razvoju $\lambda \to 0$.
- to nam da tridiagonalno matriko:
  - $A_{i,i+1} = i\,\Delta t\,\beta_s$
    $A_{i,i-1} = (i-2)\,\Delta t\,\beta_r$
    $A_{i,i} = 1 - A_{i+1,i} - A_{i-1,i}$

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    NadDia=Prepend[0]@(βsΔt Range@n0);
    PodDia=Append[0]@(βrΔt Range[0,n0-1]);
    {NadDia,1-NadDia-PodDia,PodDia}
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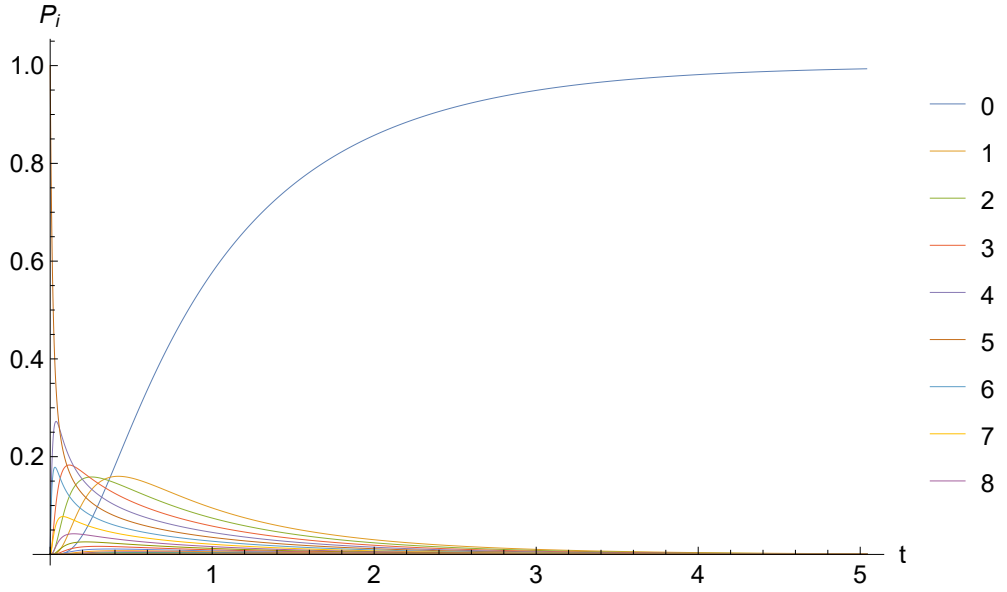
Seveda za $O(n)$ učinkovitost hranimo le naddiagonalo, diagonalo in poddiagonalo, jih po komponentah pomnožimo s $\boldsymbol{P}$ in seštejemo. To poženemo za $\beta_r = 4$, $\beta_s = 5$, $\Delta t = 0.001$, $n_0 = 250$. Tako majhen čas je potreben, saj je stvar zelo numerično nestabilna in pri veččjih korakih hitro raznese proti ∞. Dobimo naslednje spreminjanje verjetnosti:



Spreminjanje verjetnosti za različne številčnosti populacije

Vidimo, da se s časom vsa verjetnostna porazdelitev nakopiči pri 0.

Na naslednji strani je isto prikazano še s stolpci. med sličicami so enakomirni časovni koraki.

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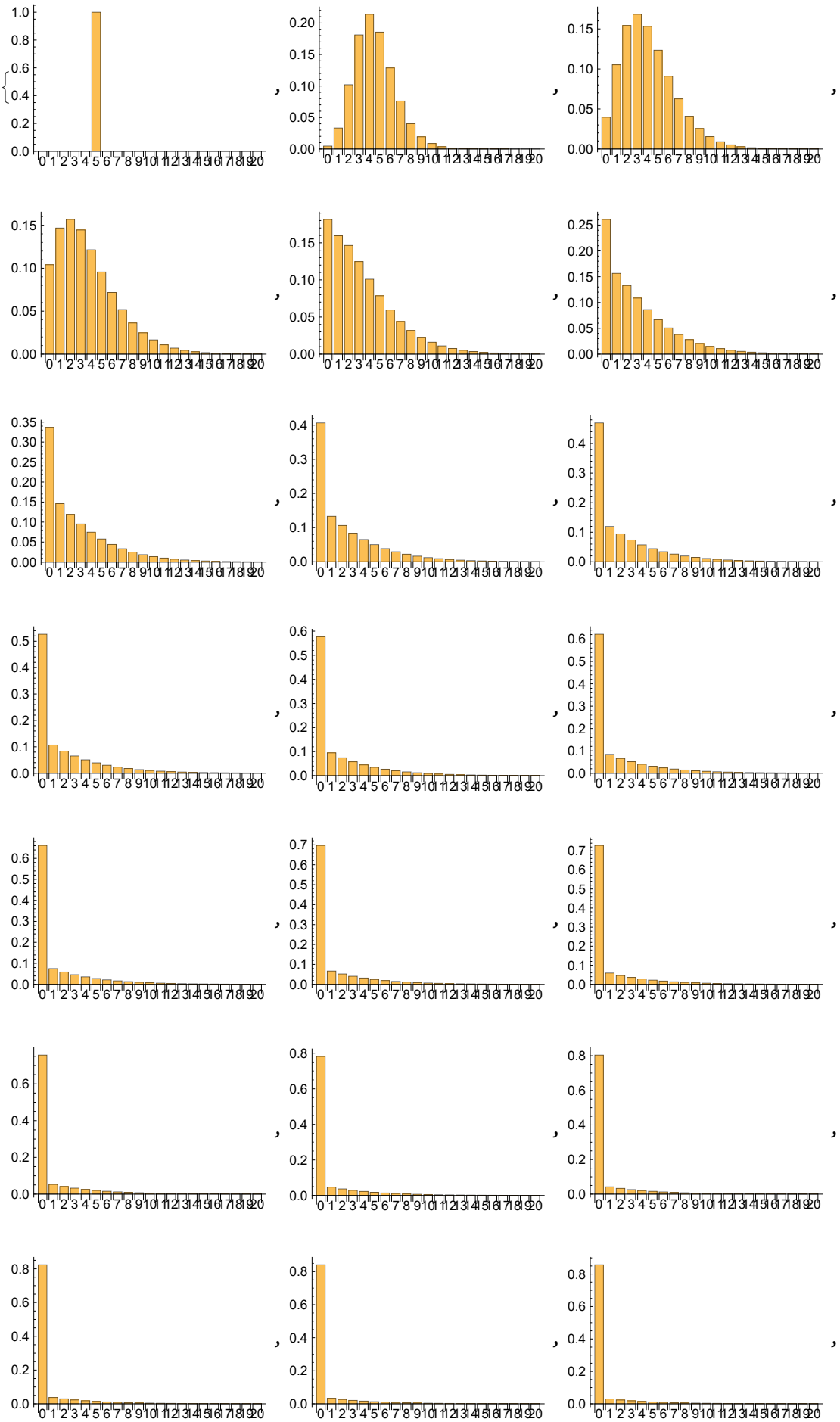

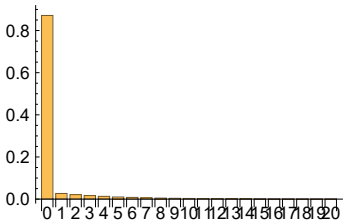
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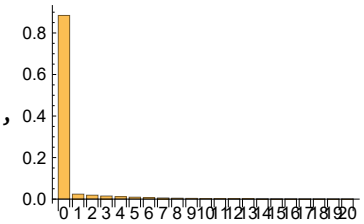
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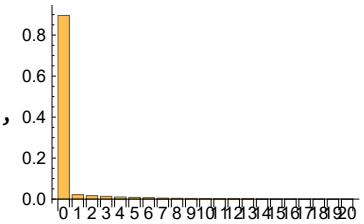
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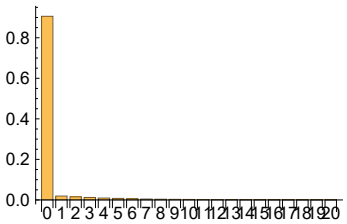
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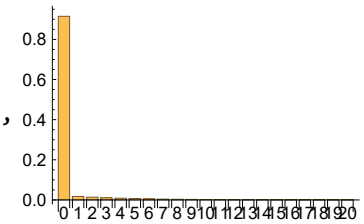
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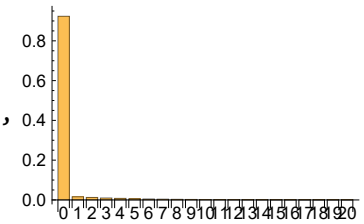
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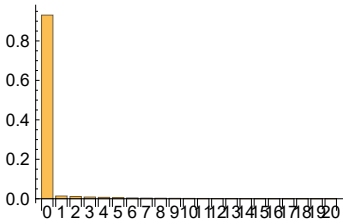
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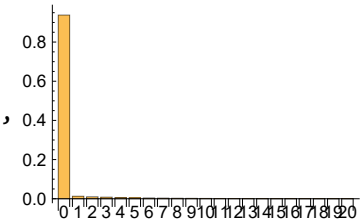
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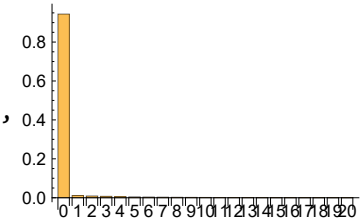
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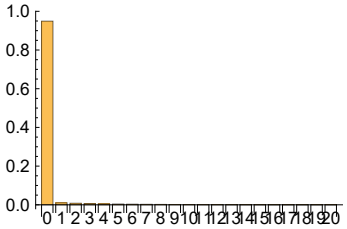
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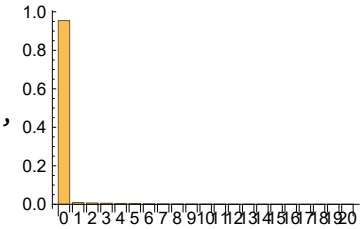
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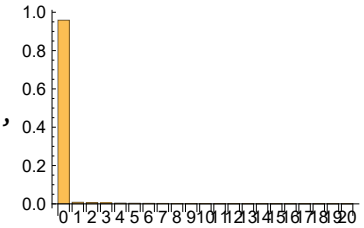
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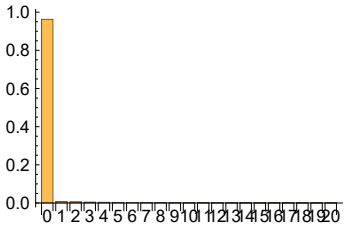
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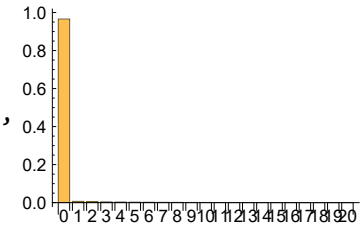
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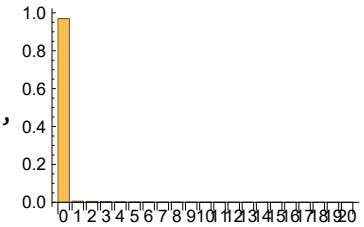
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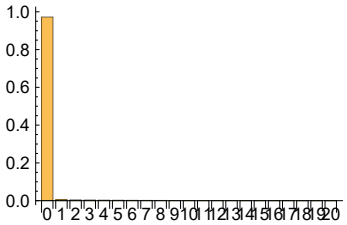
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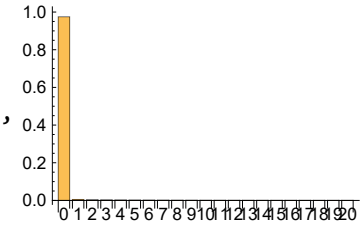
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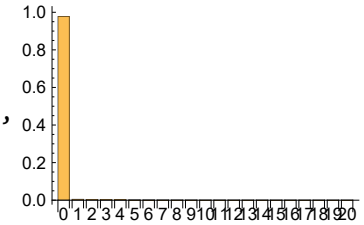
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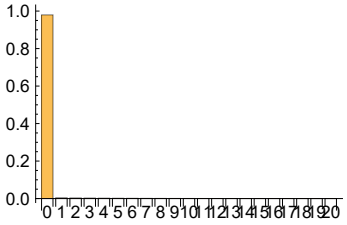
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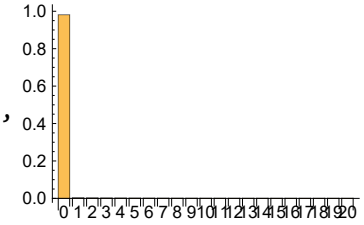
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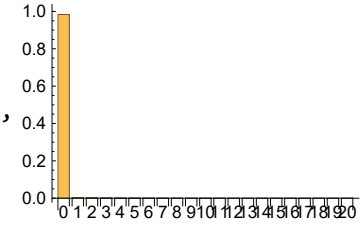
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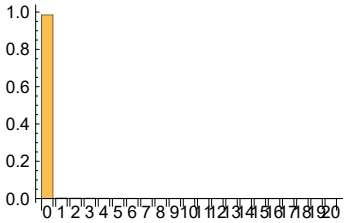

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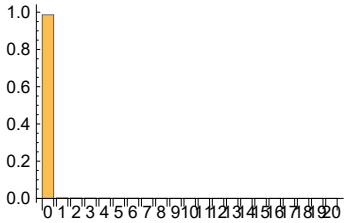

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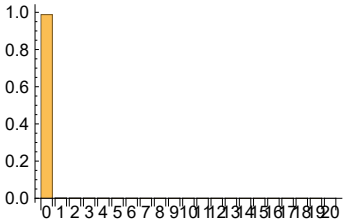

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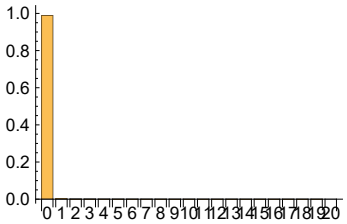

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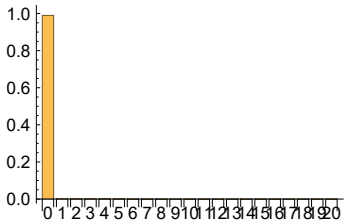

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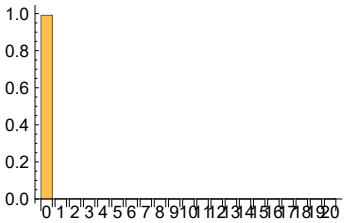

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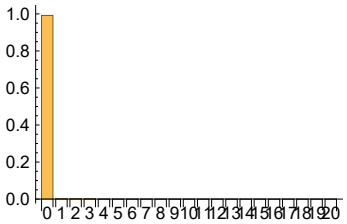

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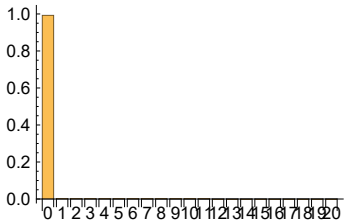

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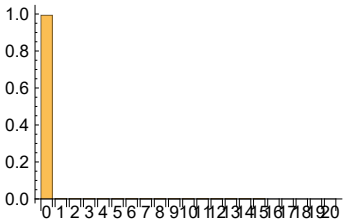


}

Čas izumrtja lahko ocenimo kot mediano: to bo takrat, ko je verjetnost v stolpcu 0 polovična. To dobimo tako, da v kubično interpolacijo vstavimo 0.5. Dobimo t_{iz} = 0.851804. To se ujema z mediano porazdelitve, ki jo dobimo kot pri 1.) nalogi. Ta pride 0.854 ± 0.003.

3.) ZAJCI-LISICE

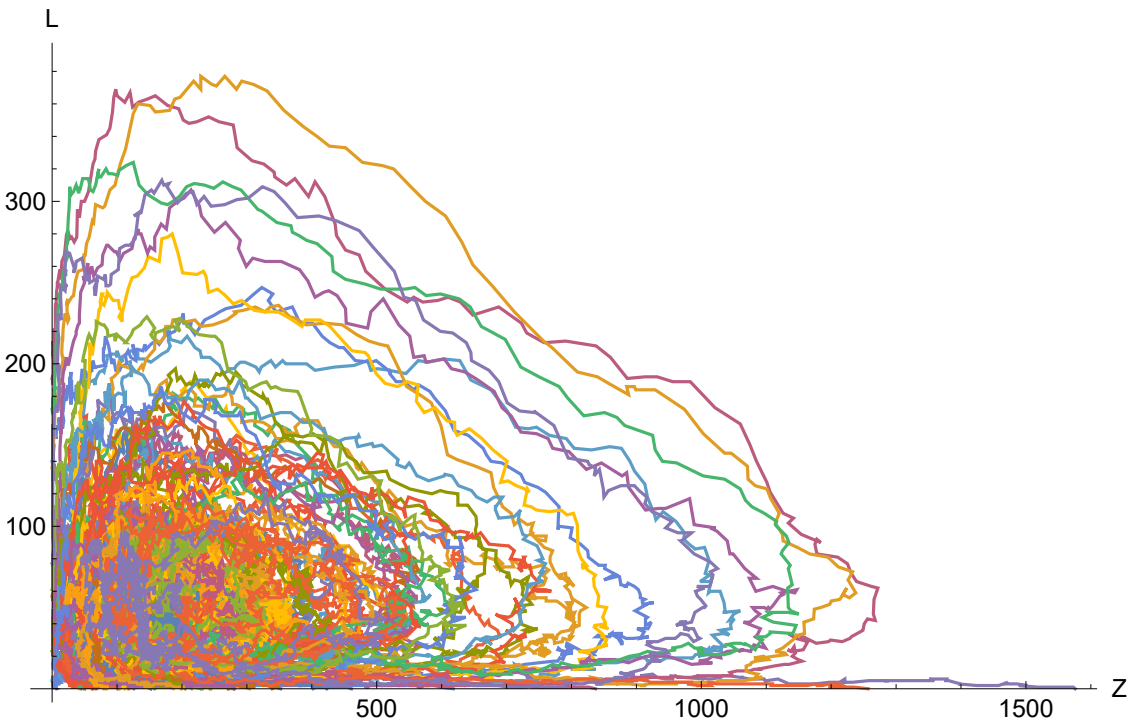
Stohastično lahko obravnavamo tudi model zajci-lisice. Dobili smo:

$$\boxed{\begin{aligned}\dot{Z} &= \rho_Z Z - \sigma_Z Z - \gamma Z L\\ \dot{L} &= \rho_L L - \sigma_L L + \delta Z L\end{aligned}}$$

Začnemo v ravnovesnem stanju z $Z_0 = 200$ zajci in $L_0 = 50$ lisicami in $\frac{\text{rodnost}}{\text{smrtnost}} = \frac{5}{4}$ za zajce in obratno za lisice. Za tako izbrane podatke dobimo stokastično rekurzijo

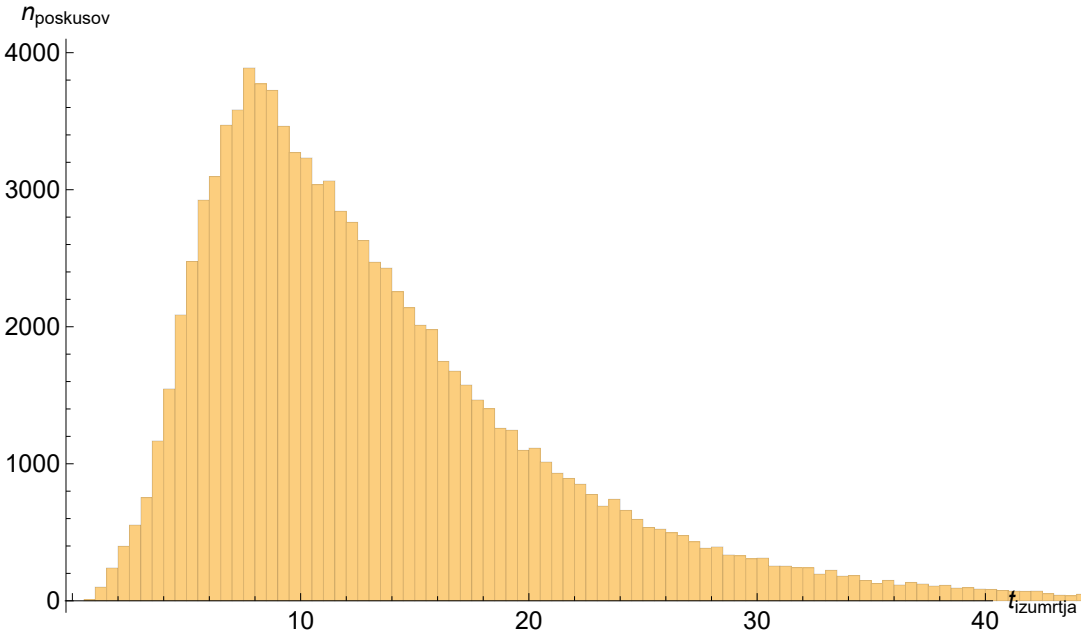
$$\boxed{\begin{aligned}Z_{n+1} &= Z_n + \mathcal{P}(5\,\alpha\, Z_n\, \Delta t) - \mathcal{P}(4\,\alpha\, Z_n\, \Delta t) - \mathcal{P}\left(\frac{\alpha}{L_0}\, Z_n\, L_n\, \Delta t\right)\\ L_{n+1} &= L_n + \mathcal{P}(4\,\beta\, L_n\, \Delta t) - \mathcal{P}(5\,\beta\, L_n\, \Delta t) - \mathcal{P}\left(\frac{\beta}{Z_0}\, Z_n\, L_n\, \Delta t\right)\end{aligned}}$$

Poženemo račun s korakom $\Delta t = .1$ in dobimo:



Stohastična dinamika zajcev in lisic

Dobimo naslednji histogram izumrtja:



Histogram časov izumrtja

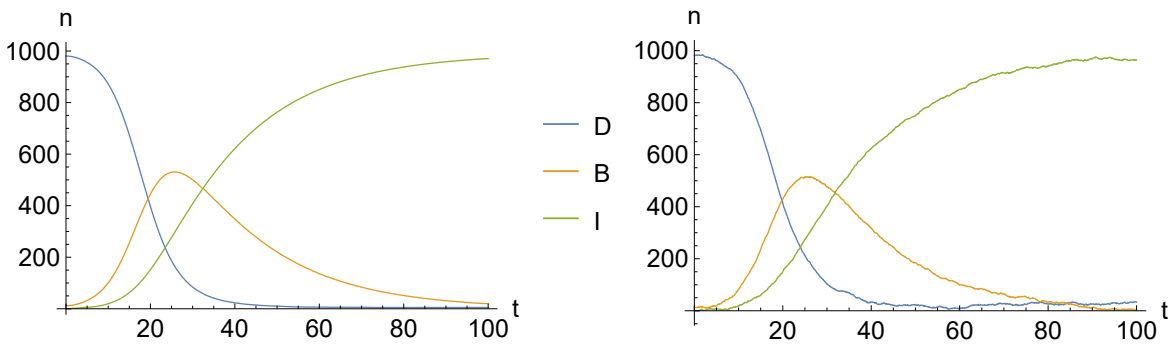
V 34% primerov najprej izumrejo lisice, v 66% pa zajci. Povprečen čas do izumrtja je 13.55.

4.) EPIDEMIJA

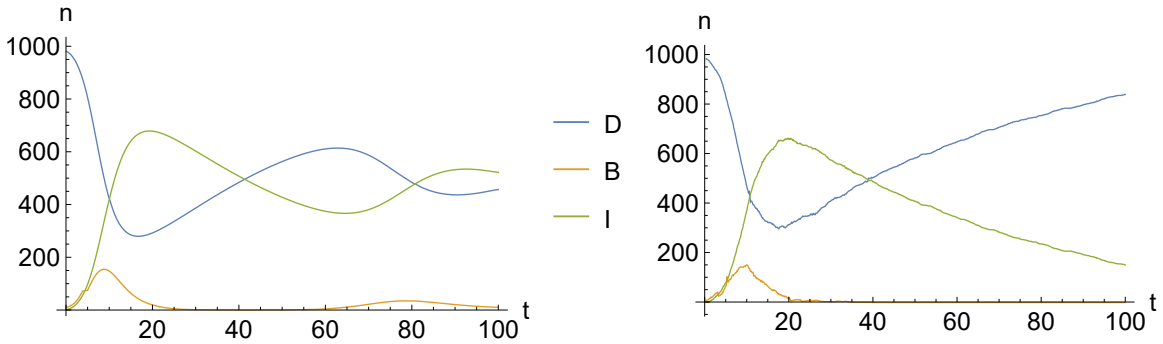
Stohastično simuliramo model epidemije iz 4. domače naloge, z dodano možnostjo, da imuni postopoma spet postajajo dovzetni. Dinamiko dovzetnih, bolnih in imunih opisuje:

$$\boxed{\begin{aligned} D' &= -\alpha D B + \gamma I \\ B' &= \alpha D B - \beta B \\ I' &= \beta B - \gamma I \end{aligned}}$$

To spet nadomestimo s poissonskim žrebom. Vzamemo korak 0.1, 998 ljudi, na začetku 15 bolnih, 0 imunih. Poglejmo 2 načilna grafa. Za $\alpha = 0.00025$, $\beta = 0.05$, $\gamma = 0$ dobimo:

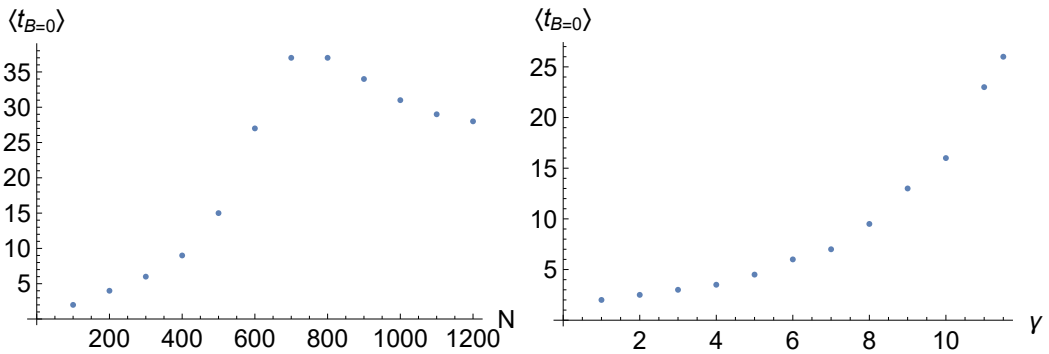


Za $\alpha = 0.001$, $\beta = 0.5$, $\gamma = 0.025$ pa:



Viimo, da število bolnih v nekem trenutku stohastično pade na 0, zato od tam naprej ni več valov epidemije, ampak le še adanje.

Poglejmo še, kako je pričakovano trajanje epidemije odvisno od števila vseh ljudi in od hitrosti izgubljanja imunosti γ.



Odvisnosti pričakovane prisotnosti bolezni od populacije N in hitrosti izgubljanja imunosti γ

Močno naraščajoča odvisnost od γ se zdi smiselna, odisnost od N pa malo preseneča, saj za več kot ~800 ljudi hitreje mine.

ZAKLJUČEK

Naučili smo se stohastično modelirati preproste dinamične sisteme. Prednost pred reševanjem diferencialnih enačb je, da lahko ocenimo koliko časa bo sistem zdržal, preden se ustavi.

KAJ BI ŠE LAHKO RAZISKALI

- še kake druge odvisnosti
- naprednejši modeli za vse 3 probleme
- laser
- nevroni: https://www.youtube.com/watch?v=gLtGVEhMFN4